Liquidity, Activism Disclosure, and Takeover Premia:
A Theory of Exit, Voice, and Corporate Control

Austin Li

University College London

Activist Returns and the Takeover Channel

- **Brav et al., 2008:** Roughly 65% of activist hedge fund returns come from M&A events
- **Greenwood and Schor, 2009:** Activist targets are acquired at significant premia; activists profit primarily through the takeover channel
- **Collin-Dufresne and Fos, 2015:** Informed trading before Schedule 13D filings moves prices — activists trade strategically on private information

Central Question

How does market liquidity shape the takeover premia that minority shareholders receive?

Three Literatures, One Missing Piece

Exit / Voice

Edmans, 2009, Maug, 1998, Admati and Pfleiderer, 2009: liquidity enables exit-based governance and stake-building for intervention.

Microstructure

Kyle, 1985, Back et al., 2018: informed trading, strategic liquidity, and price discovery.

Takeover Feedback

Edmans et al., 2012: prices guide real decisions, including takeover entry.

The Gap

No model links all three: how does market liquidity, through its effect on information in prices, shape takeover premia for minority shareholders?

Mechanism preview: Liquidity has *opposing effects*: it raises bid incidence but erodes the inference channel that sustains the activism premium. I show the net effect is non-monotone.

Disclosure Creates Two Information Regimes

Institutional facts: Schedule 13D

- Investors acquiring $>$5% stake with activist intent must file within 10 days
- Creates two observability regimes:

Quiet Voice	Public Voice
No disclosure ($D = 0$) Engagement inferred Liquidity κ matters	Filing triggers ($D = 1$) Engagement observed Liquidity irrelevant

Model Translation

- $D = \mathbf{1}\{q = +1\}$
- Buy-and-engage $\Rightarrow$ disclosure
- Hold-and-engage $\Rightarrow$ no disclosure

Disclosure shifts the economy between an *inference regime* and an *observed regime*.

Model Timeline

UCL

- **Nature** — $t=0$: v drawn; blockholder observes s
- **Trading** — $t=1$: Chooses (q, a); noise z; MM sets $P(X, D)$
- **Entry** — $t=1.5$: Bidder observes (P, D), draws ξ, entry decision
- **Payoffs** — $t=2$: Takeover or standalone payoffs

Central mechanism: Bidder entry depends on market signals, creating price-to-entry feedback.

Four Actions: Exit, Hold, Quiet Voice, Public Voice

Four possible actions (q, a):

	$a = 0$ (Passive)	$a = 1$ (Engage)
$q = -1$ (Sell)	**Exit**	—
$q = 0$ (Hold)	**Hold**	**Quiet Voice**
$q = +1$ (Buy)	—	**Public Voice**

Disclosure rule: $D = \mathbf{1}\{q = +1\}$

- Only **Public Voice** triggers disclosure
- Market must *infer* Quiet Voice from order flow

Noise trading:

- $z \in \{-1, 0, +1\}$
- $\mathbb{P}(z = 0) = 1 - \kappa$
- $\mathbb{P}(z = \pm 1) = \kappa/2$

Liquidity κ**:**

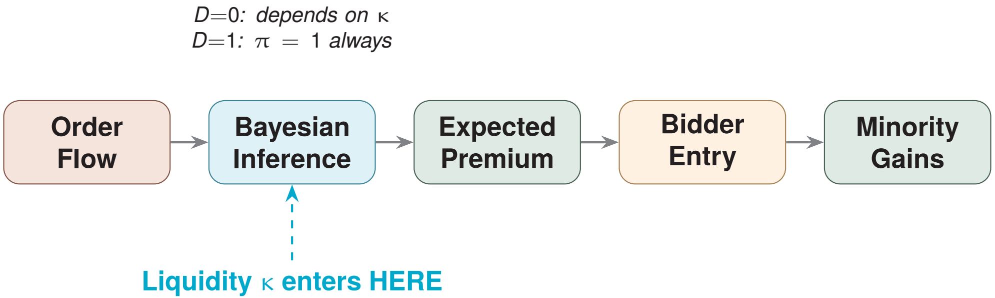
Higher κ = more noise = weaker inference.

Engagement Payoffs and Cost

$$C(s) = C_0 \exp\big(-\chi(s - \mu)/\sigma_s\big)$$

Engagement succeeds with probability ρ. If successful: standalone value $+\Delta$ and takeover premium rises from m_0 to m_1. Define expected effects: $\tilde{\Delta} \equiv \rho\Delta$ and $\tilde{m} \equiv m_0 + \rho(m_1 - m_0)$. Higher signal $\Rightarrow$ lower cost $\Rightarrow$ monotone cutoffs.

Core Mechanism: How Liquidity Affects Takeover Premia



$D = 1$ **(observed):**
$\pi(X, 1) = 1$; expected premium is $\tilde{m}$ regardless of κ — liquidity does not affect inference.

$D = 0$ **(inferred):**
$\pi(X, 0)$ depends on noise level κ — liquidity weakens the engagement signal.

Worked Inference ($D = 0$): Where Liquidity κ Enters

In the inference regime ($D = 0$), the market observes order flow $X = q + z$ but not whether the blockholder quietly engaged. Liquidity affects only the noise distribution:

$$p_0 \equiv \mathbb{P}(z = 0) = 1 - \kappa, \qquad p_1 \equiv \mathbb{P}(z = \pm 1) = \kappa/2.$$

Example: Posterior when $X = 0$

Let $\omega_E, \omega_H, \omega_Q$ denote the equilibrium probabilities of **Exit**, **Hold**, and **Quiet Voice** (from the cutoff strategy introduced next). Then $X = 0$ can arise from Hold/Quiet Voice with $z = 0$ or from Exit with $z = +1$:

$$\pi(0, 0) = \mathbb{P}(\text{Quiet Voice} \mid X = 0, D = 0) = \frac{\omega_Q \, p_0}{(\omega_H + \omega_Q)\, p_0 + \omega_E \, p_1}.$$

Key punchline: if $X = 1$ and $D = 0$ then $q = 0$ and $z = +1$, so

$$\pi(1, 0) = \frac{\omega_Q}{\omega_H + \omega_Q} \qquad \text{(independent of } \kappa\text{)}.$$

Takeaway: liquidity κ enters inference only through (p_0, p_1) and compresses posteriors toward the prior.

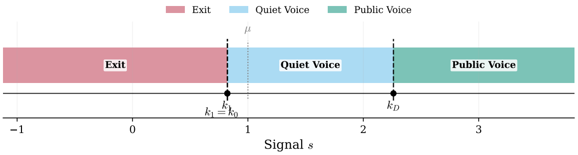
Equilibrium: Monotone Cutoff Strategy

Perfect Bayesian Equilibrium:
Monotone cutoffs $k_1 \leqslant k_0 \leqslant k_D$:

$$
\begin{aligned}
(-1, 0) &\ \text{if } s < k_1 && \text{(Exit)} \\
(0, 0) &\ \text{if } k_1 \leqslant s < k_0 && \text{(Hold)} \\
(0, 1) &\ \text{if } k_0 \leqslant s < k_D && \text{(Quiet Voice)} \\
(+1, 1) &\ \text{if } s \geqslant k_D && \text{(Public Voice)}
\end{aligned}
$$

Note: Hold and Quiet Voice produce the same order flow distribution — the market cannot distinguish them.

(Region colors above are for visual distinction; they differ from the notation color scheme.)



Equilibrium regions in (s, v) space.

Prices Embed an Activism Premium

The price embeds an activism premium through two channels:

$$P(X,D) = \delta\Big(\underbrace{(1-p^*)\,\mathbb{E}[v \mid X,D]}_{\text{fundamental value}} + \underbrace{(1-p^*)\,\tilde{\Delta}\,\pi(X,D)}_{\textbf{Standalone channel}}$$
$$+\underbrace{p^*\,(P+m_0)}_{\text{baseline takeover}} + \underbrace{p^*\,(\tilde{m}-m_0)\,\pi(X,D)}_{\textbf{Takeover channel}}\Big)$$

Standalone channel:

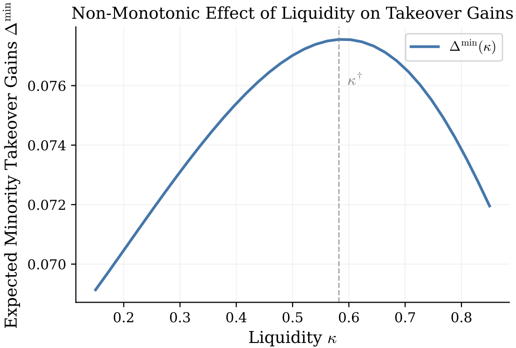
- Engagement adds $\tilde{\Delta}$ to value
- Weighted by $(1-p^*)$: conditional on no bid

Takeover channel:

- Engagement raises expected premium to $\tilde{m}$
- Weighted by p^*: conditional on bid

Key: Both channels depend on $\pi(X,D)$ — the inferred probability of engagement.

Main Result: Non-Monotone Liquidity Effect



Expected minority takeover gains:

$$\Delta^{\min}(\kappa) = m_0 \cdot \mathbb{P}(\text{bid}) + (\tilde{m} - m_0) \cdot \mathbb{E}\left[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}\right]$$

Liquidity affects (i) bid incidence through prices and (ii) bid-weighted engagement through $\pi(X, 0)$.

Why the Hump? Two Opposing Forces

Low κ (tight market)

- Strong inference $\Rightarrow$ premia track engagement accurately
- More liquidity $\uparrow$ noise cover $\Rightarrow$ bid incidence rises
- **Gains rise**

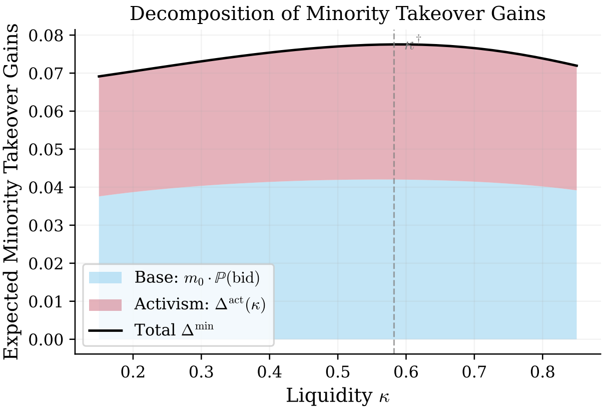
High κ (liquid market)

- Weak inference $\Rightarrow$ activism premium shrinks
- More liquidity further erodes signal
- **Gains fall**

Turning Point

At $\kappa^{\dagger}$: the marginal benefit of noise cover (more bids) equals the marginal cost of inference destruction (lower activism premium). Beyond this point, further liquidity destroys more value than it creates.

Decomposition: Base Premium vs Activism Component



Base component ($m_0 \cdot \mathbb{P}(\text{bid})$):

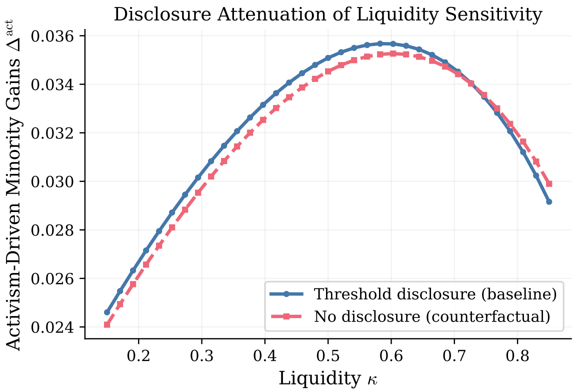
- Rises monotonically with κ
- More noise $\Rightarrow$ more bids
- Reflects bid incidence channel

Activism component:

- Hump-shaped
- Falls at high κ (weaker inference)
- Reflects composition shift in premium

Total: inherits the hump from the activism component.

Disclosure Attenuates the Liquidity Effect



$D = 1$ **states:**

- $\pi(X, 1) = 1$ always
- Expected premium $= \tilde{m}$
- Independent of κ

Sensitivity comes only from $D = 0$ **states** where $\pi(X, 0)$ varies with noise level.

Implication:
Disclosure and liquidity are **substitutes** in shaping inference — more disclosure attenuates the non-monotone liquidity effect.

Four Testable Predictions

1. **Non-monotonic liquidity–premia:** Cross-sectionally, takeover premia should be hump-shaped in stock liquidity.
   Test: quadratic liquidity term in premia regressions.
2. **Disclosure moderates:** The liquidity–premia relationship should be stronger in non-disclosed settings and muted with clear disclosure.
   Test: interact liquidity with 13D filing indicator.
3. **Resistance wedge interaction:** Where activism increases resistance, disclosure should deter bids more sharply.
   Test: triple interaction with activism type.
4. **Inference vs observability:** Activist returns should be more sensitive to market conditions in quiet campaigns than in 13D-filing campaigns.
   Test: compare returns in 13D vs non-13D activist positions.

Policy: Liquidity Regulation as Governance Policy

Disclosure threshold design:

- Lower threshold $\Rightarrow$ more $D = 1$ states
- More observability $\Rightarrow$ less inference-driven premia
- Tradeoff: transparency vs. activist incentives

Liquidity policy:

- Liquidity regulation affects corporate control, not just trading costs
- Shapes the information environment for bidder entry
- Optimal liquidity may be interior (not maximal)

Key Takeaway

Disclosure and liquidity policies are **substitutes** in shaping inference about engagement.

The SEC 13D filing window debate should account for effects on takeover premia, not just trading profits.

Summary

Mechanism

Liquidity changes what order flow reveals about hidden engagement. Disclosure creates observability vs. inference regimes. Bidder entry depends on inferred engagement through the premium wedge.

Main Prediction

Minority takeover gains are **hump-shaped** in liquidity — bid incidence rises but the activism premium composition shifts.

Contribution

Unifies exit/voice, informed trading, and takeover literatures through an inference-based mechanism.

Thank you!

`austin.li@ucl.ac.uk`

References I

- Admati, Anat R. and Paul Pfleiderer (2009). "The "Wall Street Walk" and Shareholder Activism: Exit as a Form of Voice". In: *Review of Financial Studies* 22.7, pp. 2645–2685.
- Back, Kerry, Pierre Collin-Dufresne, Vyacheslav Fos, Tao Li, and Alexander Ljungqvist (2018). "Activism, Strategic Trading, and Liquidity". In: *Econometrica* 86.4, pp. 1431–1463.
- Brav, Alon, Wei Jiang, Frank Partnoy, and Randall Thomas (2008). "Hedge Fund Activism, Corporate Governance, and Firm Performance". In: *Journal of Finance* 63.4, pp. 1729–1775.
- Collin-Dufresne, Pierre and Vyacheslav Fos (2015). "Do Prices Reveal the Presence of Informed Trading?" In: *Journal of Finance* 70.4, pp. 1555–1582.
- Edmans, Alex (2009). "Blockholder Trading, Market Efficiency, and Managerial Myopia". In: *Journal of Finance* 64.6, pp. 2481–2513.

References II

- Edmans, Alex, Itay Goldstein, and Wei Jiang (2012). "The Real Effects of Financial Markets: The Impact of Prices on Takeovers". In: *Journal of Finance* 67.3, pp. 933–971.
- Greenwood, Robin and Michael Schor (2009). "Investor Activism and Takeovers". In: *Journal of Financial Economics* 92.3, pp. 362–375.
- Kyle, Albert S. (1985). "Continuous Auctions and Insider Trading". In: *Econometrica* 53.6, pp. 1315–1335.
- Maug, Ernst (1998). "Large Shareholders as Monitors: Is There a Trade-Off Between Liquidity and Control?" In: *Journal of Finance* 53.1, pp. 65–98.

Backup: Notation Guide

Choices / Actions	
$q \in \{-1, 0, +1\}$	Trade decision
$a \in \{0, 1\}$	Engagement decision
$z \in \{-1, 0, +1\}$	Noise trade

Observables / Information	
$X = q + z$	Order flow
$D = \mathbf{1}\{q = +1\}$	Disclosure flag
s	Private signal
v	Fundamental value

Equilibrium Outcomes	
$P(X, D)$	Price
$\pi(X, D)$	Posterior engagement prob.
$p(P, D)$	Bid probability
$m(X, D)$	Expected premium

Parameters	
κ	Liquidity (noise prob.)
k_1, k_0, k_D	Signal cutoffs
ρ	Success prob.
m_0, m_1	Base / success premium
$\tilde{m}$	Expected premium
$\Delta, \tilde{\Delta}$	Value-add; $\tilde{\Delta} = \rho\Delta$

Color coding: Blue = choices/actions Red = observables Green = outcomes

Backup: Baseline Calibration

Baseline calibration	
Discount factor δ	0.95
Liquidity κ	0.50
Engagement: (C_0, χ)	(0.12, 0.50)
Engagement: (Δ, ρ)	(0.25, 0.90)
Resistance: (m_0, m_1)	(0.10, 0.30)
Baseline synergy $\bar{S}$	1.44
Entry cost / shock (K, σ_ξ)	(0.15, 0.40)
Signal noise $(\sigma_v, \sigma_\varepsilon)$	(0.50, 0.50)

Cutoffs: $k_1 = k_0 \approx 0.82$, $k_D \approx 2.26$.

D	Order flow X	$P(X, D)$	$\pi(X, D)$	$p(X, D)$	$m(X, D)$
0	-2	0.84	0.00	0.81	0.10
0	-1	1.06	0.41	0.55	0.17
0	0	1.23	0.74	0.33	0.23
0	1	1.39	1.00	0.17	0.28
1	0	1.90	1.00	0.01	0.28
1	1	1.90	1.00	0.01	0.28
1	2	1.90	1.00	0.01	0.28

- Disclosed states ($D = 1$): $\pi = 1$, nearly flat prices
- Bids are rarer when $D = 1$ due to higher resistance wedge

Backup: Full Posterior Formulas

With stake disclosure $D = \mathbf{1}\{q = +1\}$:

Disclosed branch ($D = 1$):

- $X \in \{0, 1, 2\}$ (since $q = +1$)
- $\pi(X, 1) = 1$ for all X
- Engagement fully observed

Let $p_0 = 1 - \kappa$, $p_1 = \kappa/2$:

Non-disclosed branch ($D = 0$):

- $X \in \{-2, -1, 0, 1\}$ (since $q \in \{-1, 0\}$)
- $\pi(X, 0)$ via Bayes' rule
- Depends on κ through (p_0, p_1)

$$\pi(1, 0) = \frac{\omega_Q}{\omega_H + \omega_Q}$$

$$\pi(0, 0) = \frac{\omega_Q p_0}{(\omega_H + \omega_Q)p_0 + \omega_E p_1}$$

$$\pi(-1, 0) = \frac{\omega_Q p_1}{(\omega_H + \omega_Q)p_1 + \omega_E p_0}$$

Note: $\pi(1, 0) = \omega_Q/(\omega_H + \omega_Q)$ does *not* depend on κ because $X = 1$ with $D = 0$ pins down $q = 0$.

Backup: Solving the Pricing Fixed Point

For each information set (X, D):

1. Compute $\pi(X, D)$ from Bayes' rule
2. Compute $\hat{V}(X, D) = \mathbb{E}[v \mid X, D] + \tilde{\Delta}\,\pi(X, D)$
3. Solve scalar fixed point:

$$P = \delta\big((1 - p(P, D))\,\hat{V} + p(P, D)(P + m(X, D))\big)$$

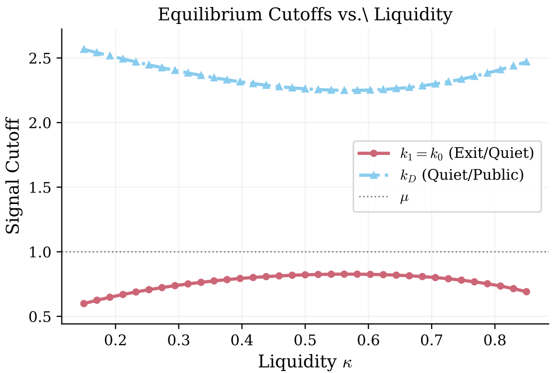
   where $p(P, D) = 1 - \Phi\left(\frac{m(X,D)+K-\bar{S}+P}{\sigma_\xi}\right)$

   (We write $m(X, D)$ since $P(\cdot, D)$ is injective, so the bidder infers X from P.)

4. Under the regularity condition (moderate feedback), the map is a contraction; iterate to convergence

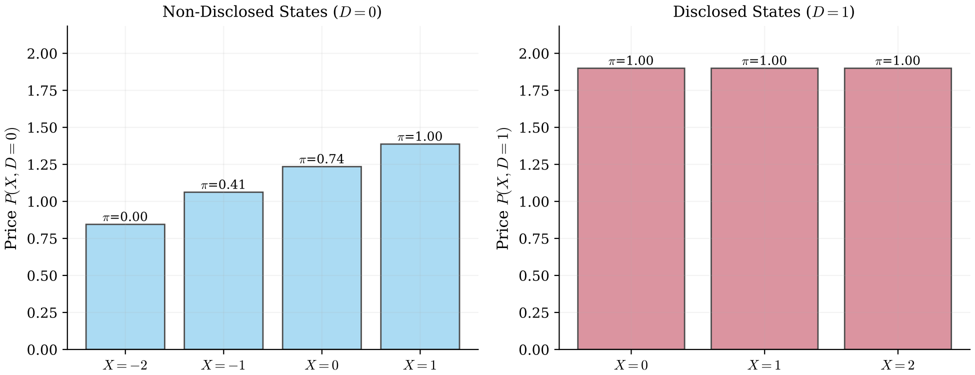
Regularity condition: The derivative of bid probability w.r.t. price is bounded, ensuring a unique fixed point. Specifically, $\delta\,\phi(0)/\sigma_\xi < 1$.

Backup: Cutoffs vs Liquidity



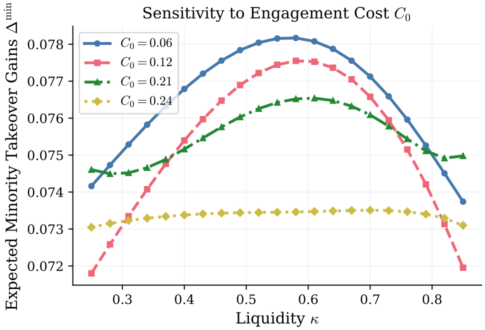
- Liquidity shifts Exit/Voice boundaries
- Weak ordering $k_1 \leqslant k_0 \leqslant k_D$ permits region collapse
- Exit becomes more attractive as κ rises (noise cover for selling)

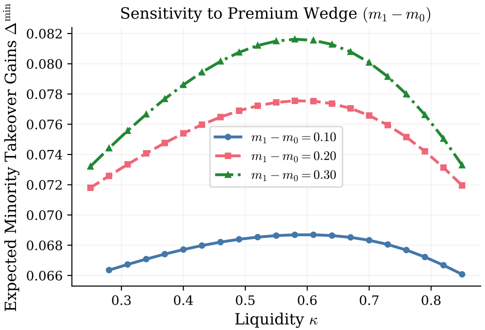
Backup: Price Function



$D=0$ **(non-disclosed):** Prices vary with X as the market infers engagement probability from order flow.

$D=1$ **(disclosed):** Prices are high and nearly flat across X; $\pi=1$ ensures full activism premium.





Varying engagement cost C_0

Higher C_0: shrinks voice regions, lowers activism premia, but can raise bids by lowering resistance expectations.

Varying resistance wedge $m_1 - m_0$

Larger wedge: raises premia conditional on bid but deters entry through higher expected resistance.

Backup: Alternative Disclosure Regimes (FI / ND / NR)

Replace (X, D) by (X, S) with public signal S; define $\pi(X, S) = \mathbb{P}(a = 1 \mid X, S)$.

- **FI (Full Information):** $S_{\text{FI}} = (q, a)$ reveals action $\Rightarrow$ inference channel shut down
- **ND (No Disclosure):** $S_{\text{ND}} = \emptyset \Rightarrow$ engagement inferred from X in *all* states
- **NR (Noisy Rumor):** $S_{\text{NR}} = (D, R)$ adds a rumor R informative about quiet engagement when $D = 0$

X	$\pi(X, 0)$	$\pi_{\text{ND}}(X)$	$\pi_{\text{NR}}(X, 0, R = 0)$	$\pi_{\text{NR}}(X, 0, R = 1)$
-1	0.41	0.41	0.19	0.68
0	0.74	0.74	0.48	0.89
1	1.00	1.00	1.00	1.00

(q, a)	π_{FI}
$(-1, 0)$	0.00
$(0, 0)$	0.00
$(0, 1)$	1.00
$(1, 1)$	1.00

Message: Disclosure informativeness/timing reshapes $\pi(\cdot)$ and hence the liquidity–premia mapping.